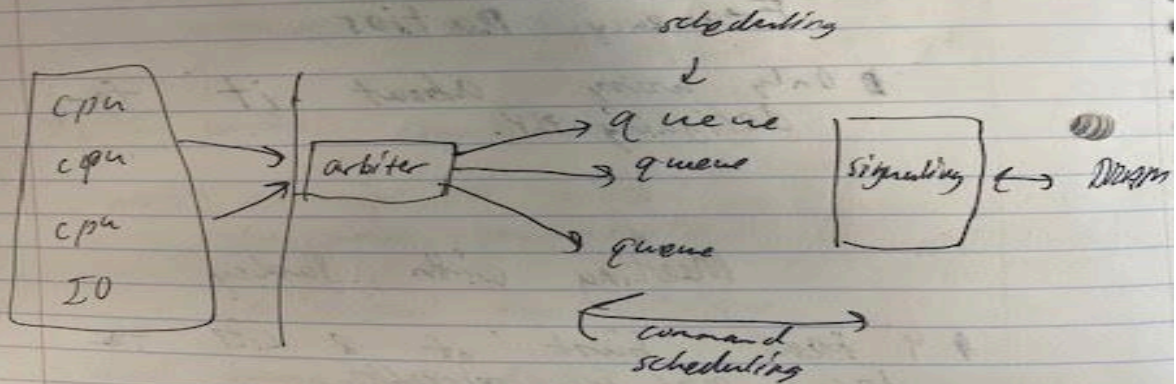
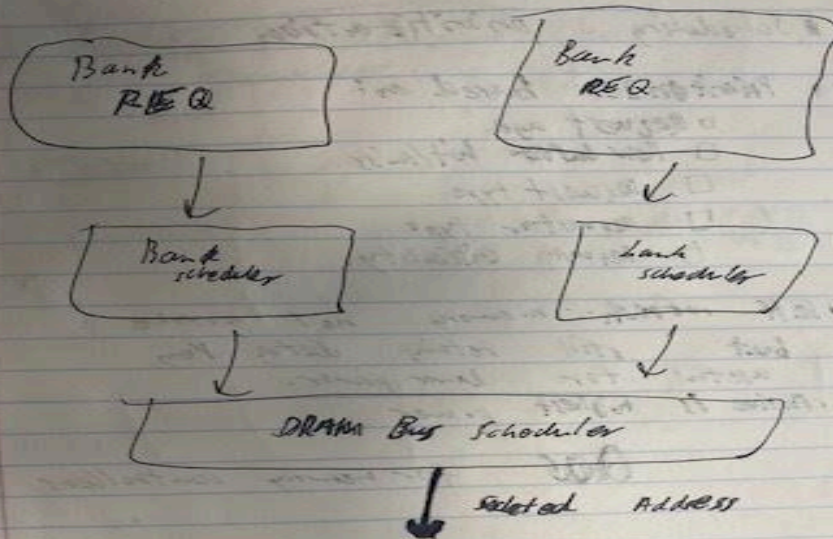


DRAM Controller Lecture

- Follow timing constraints.
- Buffer and schedule requests.
- Manage power in LPDDR





• Scheduling policies

0 First come first serve, bank

• FCFS

1. Row hit

2. Select first

maximize hit rate and maximize throughput.

•

column commands prioritized over row commands. older ones first.

for
memory

Scheduling prioritization

Prioritize based on:

- ☐ Request age
- ☐ Row buffer hit/miss
- ☐ Request type
- ☐ Requestor type
- ☐ Request criticality.

- Self refresh: memory not active but still retains data. Very useful for lower power.
- Active is highest power

QoS for memory controllers